Documentation to the L0PRS program

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This outline documents a simple implementation of a polygenic risk score (PRS) for GWAS data, that is, case-control data genotyped at a number of genetic variants, SNPs. It makes use of the `--score` function in *plink* [1-3], based on an allelic association test, `--assoc`. A PRS is computed for each case and control individual, with cases on average showing higher scores than controls. Classification (phenotype prediction) is performed by calling an individual a "case" if the individual's score is above the 99th percentile of control scores and otherwise is called a "control". This program is analogous to *L1outPRS* except that no cross-validation is performed, which makes *L0PRS* run much faster than *L1outPRS*. The Pascal source code was compiled with the *fpc* package in Ubuntu (https://www.freepascal.org/). The current version is dated 25 Oct 2026.

The *L0PRS* program will compute a PRS for each of the best 3, 5, 10, 20, …, 50,000 variants (14 such sets of variants), where "best" refers to the ordering of variants selected by command line parameter #2 (see below). For each such selection, predicted phenotypes will be computed.

Once a prediction has been made for each individual, the following parameters are in general use:

Known phenotype	Predict "case"	Predict "control"
case	*a*	*b*
control	*c*	*d*

The letters, *a*, *b*, *c*, and *d*, indicate numbers of individuals, for example, *a* = number of cases predicted to be cases. Sensitivity, or power, of a test to identify cases is given by the True Positive Rate, TPR = $a/(a + b)$, and the False Positive Rate is FPR = $c/(c + d)$; positive predictive value, PPV = $a/(a + c)$; negative predictive value, NPV = $d/(b + d)$; odds ratio, OR = $(a \times d)/(b \times c)$; prediction accuracy, ACC = $(a + d)/(a + b + c + d)$.

Both, OR and ACC, indicate prediction accuracy but on different scales. For the special situation, PPV = NPV, we can write OR = $[ACC/(1 - ACC)]^2$, which demonstrates that OR increases with ACC.

Included is a sample case-control dataset, AMDHKg, on wet age-related macular degeneration (AMD), collected in Hong Kong and published in 2006 [5]. The significant SNP in that publication is `rs10490924` on chromosome 10.

Running the *L0PRS* program

This program creates an allelic polygenic risk score (PRS) as implemented in *plink 1.9* with the allelic scoring function, `--score`. Because *L0PRS* uses the Linux `sort` function (which is very different from `sort` in Windows!), it runs properly only in Linux, for example, Ubuntu (Linux executable = `L0PRS`).

Step 1. Download *plink* version 1.9 for Linux from https://www.cog-genomics.org/plink/ .

Step 2. Make the *plink* file executable, `chmod +x plink`, and rename it, `mv plink plink19`.

Step 3. Copy the *plink19* file to a specific folder in the program path: `sudo cp plink19 /usr/local/bin`. Analogously, make the program file executable, `chmod +x L0PRS`, and copy it to a suitable folder, `sudo cp L0PRS /usr/local/bin`.

Step 4. Make a suitable work directory and copy a *plink* formatted file set, for example AMDHKg.map and AMDHKg.ped, into this directory. On the console, type `L0PRS` followed by command line parameters.

Command line parameters are as shown below. You may also see these by just typing the program name, `PRS`, without any trailing parameters.

1) The *plink* fileset name, without ".map" and ".ped", for example, AMDHKg. If problems occur down the line, apply a minor allele frequency of 0.001 or more and run the program again.
2) A number, 1 or 2, to indicate whether the variants are to be ordered by the *p*-value (by 1 – *p*, actually) (1) or by the odds ratio (2) (2 = normal use).
3) A number, 1 or 2, to use prediction accuracy, ACC (1), or the positive predictive value, PPV (2), as the score variable (1 = normal use).

Optionally:

4) A number, *n*, indicating that calculations should only be carried out for the best *n* variants, that is, for a single category of *n* SNPs, where *n* is preferably a number between 1 and 50,000. Output will be, for each individual, the observed phenotype, the predicted phenotype, and the risk score for that individual. Also, the log file will contain the AUC/AUROC value.
5) Number of FPR classes for calculation and graphing of AUC.

An example of a command-line job submission is as follows:

```
jurg@Lap2023:~$ L0PRS AMDHKg 2 1
```

Preferably start such a job in the foreground to see potential error messages and verify that everything is going well. If so, interrupt the job after a minute or two and submit it to run in the background:

```
jurg@Lap2023:~$ (L0PRS AMDHKg 2 1 >/dev/null)&
```

To run only one set of variants, for example, the 1,200 variants with largest OR values and 10 classes of AUC, type

```
jurg@Lap2023:~$ L0PRS AMDHKg 2 1 1200 10
```

or

```
jurg@Lap2023:~$ (L0PRS AMDHKg 2 1 1200 10 >/dev/null)&
```

A report file, `L0PRS.AMDHKg-21.rpt`, will be created in the directory above the running directory and will contain information on numbers of replicates processed. To quickly see output of the report file from the running directory, type `cat ../*rpt`. The four numbers, *a* … *d*, in the output form a 2 × 2 table, which will be analyzed with the Fisher exact test. You may want to delete this file after your run has completed.

Currently the following maximum program constants apply:

50,000 individuals
1,000 classes of AUC values

If *plink* issues many notes (one or two such notes might be ok) saying that a line in a score file was omitted because of allele incompatibility, try using the same dataset but with increased MAF.

Comparing parameter settings

Consider the following idealized conditions:

- The number of cases to controls is in the ratio 1:r (there are r times as many controls as cases)
- There is a threshold (percentile), t, of controls, and individuals with a score above t are predicted to be cases, otherwise they are predicted to be controls
- A proportion, s, of cases have scores below t and are, thus, predicted to be controls; otherwise, they are predicted to be cases.

Then we can compute the resulting prediction parameters as shown in the tables below, where accuracy refers to the proportion of correctly classified individuals.

Known phenotype	Predict "case"	Predict "control"	Sum
cases	$(1 - s)n$	sn	n
controls	$(1 - t)rn$	trn	rn

t = 99th percentile of controls	r = 1			r = 10		
	s = 0.10	s = 0.30	s = 0.50	s = 0.10	s = 0.30	s = 0.50
Accuracy, ACC	0.95	0.85	0.75	0.98	0.96	0.95
Positive predictive value, PPV	0.99	0.99	0.98	0.90	0.88	0.83
Negative predictive value, NPV	0.91	0.77	0.66	0.99	0.97	0.95

Clearly, these parameters can tolerate quite a bit of overlap between cases and controls. For ACC and NPV, more controls than cases is better than equal numbers, but the opposite is true for PPV.

Computing AUC

With optional 4th and 5th command line parameters, the *L0PRS* program will prepare a list of N control PRS scores, s, sorted from small to large, where N is the number of control individuals. A threshold, t, is iteratively assumed as the average between two successive control scores, that is, $t_1 = (s_1 + s_2)/2$, $t_2 = (s_2 + s_3)/2$, …, $t_{N-1} = (s_{N-1} + s_N).2$. For a given threshold, individuals with scores below it are classified as controls, and we can count how many of them are known cases and controls; analogously, individuals with scores above the threshold are classified as cases. Thus, for each of the $N - 1$ thresholds, we get four numbers, a, b, c, and d, from which we compute FPR and TPR. The collection of these pairs of false and true positive rates allows us to draw an ROC (Receiver Operating Characteristics) curve and the area, AUC, under it.

The program will then write 3 output files, notably, `L0PRS.s.AMDHKg.1200_SNPs-21-100.txt`, indicating that 1,200 SNPs were run and 100 classes of FPR/TPR pairs of values were formed. This output file contains the following three sections:

1. **List of individuals**. For each individual, the known and predicted phenotypes are provided along with that individual's score value. Also, the 99th %ile of control scores is listed (constant for all individuals).
2. **List of assumed control thresholds**. For each such assumed threshold, the associated prediction tables and their FPR and TPR values are presented.
3. **AUC graph**. Classes of FPR and TPR are given, which may be used to draw an ROC curve. Such a curve may also be prepared based on all FPR-TPR values provided in the previous section. This section ends with the calculated AUC value (based on FPR classes) and a number, n_0, indicating the number of classes without known FPR/TPR values. If $n_0 > 0$, you may want to do another run with a smaller number of FPR classes so that $n_0 = 0$.

References

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